

Management of Fever in the Immunocompromised Rheumatology Patient

A) Inclusion Criteria

Rheumatologic patient with $T_{\max} \geq 38.5$ degree Celsius who is immunosuppressed, i.e. any one of the following:

- Immunosuppressive medications^a
 - Any biologic (e.g. anti-TNF-alpha, anti-IL1, anti-IL6, anti-CD20 etc.)
 - Mycophenolate
 - Azathioprine
 - Cyclophosphamide and other alkylating agents within the last 3 months
 - Calcineurin inhibitor e.g. tacrolimus, ciclosporin
 - Systemic corticosteroids
 - Pulsed IV methylprednisolone within last 3 months
 - Prednisolone dose of 1 mg/kg daily or 10 mg daily or equivalent
 - Methotrexate $\geq 0.5\text{mg/kg/week}$
- Certain rheumatological diseases when uncontrolled, can in itself lead to a suppressed immune system; e.g. systemic lupus erythematosus (SLE). If this is a concern, discuss with the Paediatric Rheumatologist on call.

Patients who satisfy the above should be triaged as **P2** or, if haemodynamically unstable, **P1**.

^aMedications that are not considered immunosuppressive:

- Hydroxychloroquine
- Sulphasalazine
- Leflunomide
- Colchicine
- Intravenous immunoglobulin (IVIG)

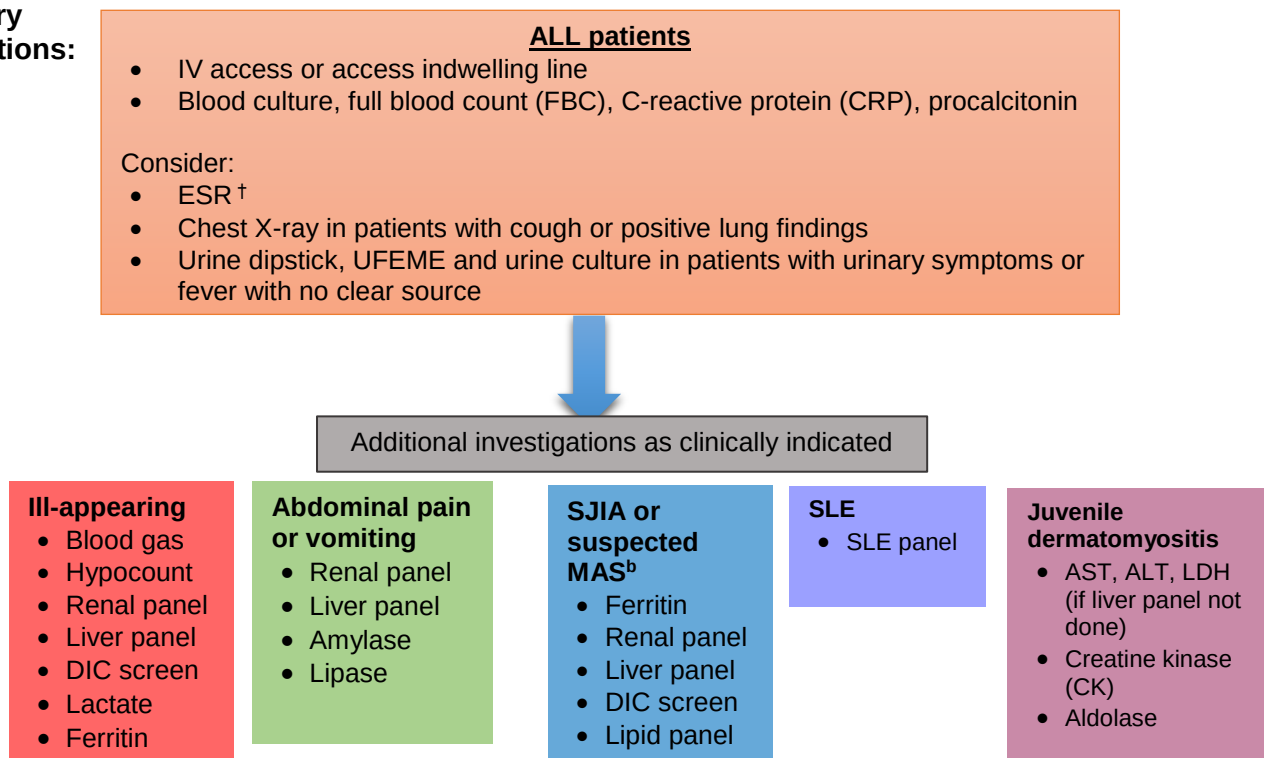
B) Assessment

- Assess if acute resuscitation is required
- Obtain history and physical examination to evaluate cause of fever – infection or disease flare.
- Infection history and previous positive cultures

C) Investigations

Disclaimer: The following flow chart is not applicable for patients with a clear viral aetiology e.g. varicella, zoster, hand, foot and mouth disease. The Paediatric Rheumatologist on-call should be contacted to discuss the management of these patients.

Laboratory investigations:



[†]Procalcitonin and CRP are specific but not sensitive for bacterial infections, i.e. elevated levels increases the likelihood of a bacterial infection, however, normal values do not exclude it.

^bMacrophage activation syndrome (MAS) is a complication associated with active systemic juvenile idiopathic arthritis and other active rheumatological conditions such as SLE, juvenile dermatomyositis, Kawasaki's disease.

Suspect macrophage activation syndrome (MAS) if a child has a rash, generalized lymphadenopathy, and hepatosplenomegaly on clinical examination. They may present with CNS dysfunction or haemorrhagic manifestations as well.

Typical laboratory features include cytopenias, transaminitis, hyperferritinaemia, raised D-dimer and hypofibrinogenaemia.

D) Management

i) Antibiotics

- If patient is **ill-appearing or neutropenic** ($ANC < 1.0 \times 10^9/L$), **intravenous broad spectrum antibiotics should be administered within 60 minutes of arrival.**
- Choice of antibiotics:
 - Ill-appearing or neutropenic: IV cefepime
 - All other patients: IV ceftriaxone

ii) Consultation with Paediatric Rheumatologist on call

- Consult paediatric rheumatologist on call if:
 - Patient is well, non-neutropenic and has no clear indication for inpatient intravenous antibiotics (e.g. upper respiratory tract infection, hand, foot, and mouth disease)
 - Disease flare suspected
 - Macrophage activation syndrome (MAS)
- Discuss with the paediatric rheumatologist on call **prior to discontinuation** of any immunosuppression during period of illness.

iii) Risk of adrenal insufficiency – need for steroids

- Corticosteroid therapy should **not** be stopped due to risk for adrenal insufficiency. In the event that oral corticosteroid is not tolerated, equivalent doses of intravenous hydrocortisone should be given.
- **Stress doses of corticosteroid** (IV hydrocortisone 100 mg/m² loading followed by 100 mg/m²/day, given 6 hourly) should be administered for patients with fluid-refractory and catecholamine-refractory septic shock requiring initiation of a second inotrope.

iv) Disposition

- Admit all patients unless otherwise discussed with the paediatric rheumatologist on call.
- Upon agreement with the paediatric rheumatologist on-call, well-appearing patients may be sent home with oral augmentin, after one dose of IV ceftriaxone is given in Children's Emergency. They should be given a review appointment with paediatric rheumatology within 2 days.

References:

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Summary of management of fever in immunocompromised rheumatology patient

